



MACHINE LEARNING

CONFERENCE STYLE PAPER

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A Machine Learning Approach to Telecom Customer Churn Prediction

Abstract

Customer churn prediction is a critical task in the telecommunications industry, where retaining existing customers is more cost-effective than acquiring new ones. This study develops a machine learning pipeline to predict customer churn using a real-world dataset consisting of customer demographics, service usage, and billing information.

Customer churn predictions are widely studied in machine learning due to their impact on business profitability [1].

The workflow includes data preprocessing, exploratory data analysis, feature engineering, and model development. Class imbalance was addressed using SMOTE, and feature scaling was applied where necessary. Multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM), were implemented and evaluated. Hyperparameter tuning was performed using GridSearchCV to improve model generalisation.

The results indicate that Random Forest achieved the best performance after tuning, with an accuracy of 81%. The study highlights the importance of structured preprocessing and proper evaluation in building reliable machine learning systems.

Keywords

Customer Churn, Machine Learning, SMOTE, GridSearchCV, Classification, Feature Engineering

1. Introduction

Customer churn prediction is one of the most important applications of machine learning in

industries where customer retention directly influences revenue. In highly competitive sectors such as telecommunications, customers can easily switch service providers, making it essential for companies to identify potential churners in advance.

The primary objective of this project is to develop a machine learning-based solution capable of predicting whether a customer is likely to churn. By analysing customer demographics, service usage, and billing information, predictive models can assist businesses in implementing targeted retention strategies.

This study adopts a structured machine learning pipeline, starting from data preprocessing and exploratory data analysis to model development and evaluation. Multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM), are implemented and compared. Special emphasis is placed on handling class imbalance, preventing overfitting, and ensuring that models generalise well to unseen data.

By evaluating and comparing different models, this project aims to identify the most effective approach for churn prediction while also highlighting the practical challenges encountered during implementation

2. Dataset Description

The dataset used in this study contains **7043 customer records with 37 features**, including:

- Demographic details (Age, Gender, Dependents)
- Service usage (Internet Service, Contract Type)

- Financial data (Monthly Charges, Total Revenue)

The dataset contains a mixture of categorical and numerical features, which required careful preprocessing before model implementation. Features such as “Internet Service”, “Contract Type”, and “Payment Method” provide insights into customer behaviour, while numerical attributes such as “Tenure in Months” and “Monthly Charges” help capture usage patterns.

One important consideration during preprocessing was the presence of features that could lead to data leakage. Variables such as “Churn Category” and “Churn Reason” directly reveal the outcome and were therefore removed to ensure that the models learn from meaningful predictors rather than future information.

Additionally, maintaining the original dataset size was crucial. Instead of dropping rows with missing values, appropriate handling techniques were applied to preserve the integrity of the dataset and ensure sufficient data for training.

Preprocessing Decisions:

Removed:

- Customer ID (non-informative)
- Churn Category & Reason (data leakage risk)
- Customer Status (redundant with target)

Care was taken to avoid unnecessary row deletion during missing value handling to preserve dataset integrity.

Final dataset: 7043 rows x 31 features

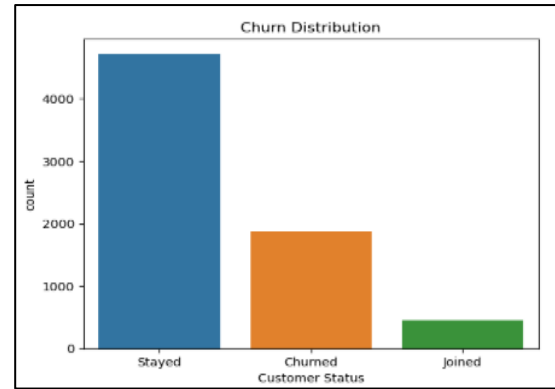


Figure 1: Distribution of Customer Churn Before SMOTE

As shown in Figure 1, the dataset is imbalanced, with significantly more non-churn customers than churn customers.

3. Experimental Setup

The project followed a structured pipeline:

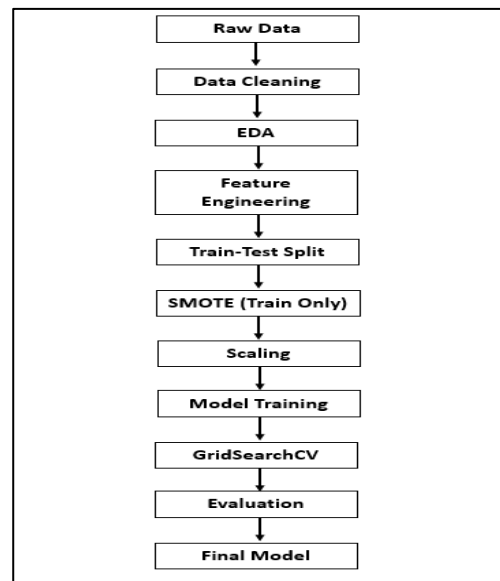


Figure 5: Machine Learning Pipeline Architecture

A structured pipeline was followed to ensure consistency and reproducibility throughout the project. Each stage of the workflow was carefully designed to address common machine learning challenges such as data imbalance, feature scaling, and model overfitting.

The dataset was split into training and testing sets to evaluate model performance on unseen

data. SMOTE was applied only to the training set to balance the class distribution, ensuring that the test set remained representative of real-world scenarios. Feature scaling was applied where necessary, particularly for distance-based algorithms such as KNN and SVM.

This systematic approach ensured that all models were trained and evaluated under consistent conditions, allowing for a fair comparison of their performance.

4. Methods

Models Used:

- Logistic Regression (baseline + interpretable)
- Decision Tree (simple tree-based model)
- Random Forest (ensemble method)
- KNN (distance-based)
- SVM (margin-based classifier)

Justification:

Each model was selected to represent a different learning approach, allowing a comprehensive comparison of performance across different algorithm types.

The implementation was carried out using Python and the Scikit-learn library [2].

Each model used in this study represents a different approach to classification. Logistic Regression serves as a baseline model due to its simplicity and interpretability. Decision Trees provide a rule-based structure that is easy to understand but can be prone to overfitting. Random Forest improves upon Decision Trees by combining multiple trees, reducing variance and increasing robustness.

K-Nearest Neighbours is a distance-based algorithm that classifies data points based on similarity, making it sensitive to feature scaling. Support Vector Machines aim to find an optimal decision boundary that maximises the

margin between classes, but they can be computationally expensive for larger datasets.

5. Exploratory Data Analysis

EDA revealed several key patterns:

- Customers with lower tenure were more likely to churn
- Higher monthly charges were associated with churn
- Contract type significantly influenced customer retention

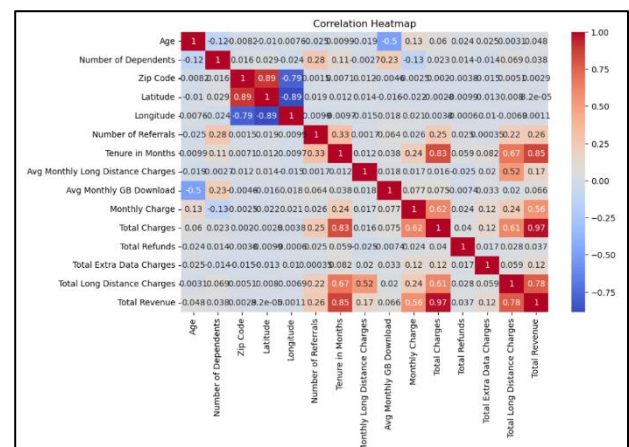


Figure 2: Correlation Heatmap of Numerical Features.

Exploratory Data Analysis played a crucial role in understanding the underlying patterns within the dataset. It provided insights into how different features influence customer churn and guided the feature selection process.

For instance, customers with shorter tenure were observed to have a higher likelihood of churn, indicating that customer loyalty increases over time. Similarly, higher monthly charges were associated with increased churn, suggesting that pricing may influence customer decisions.

These insights not only helped in feature selection but also provided a better understanding of the problem domain.

6. Feature Engineering

Since the dataset contained a combination of categorical and numerical variables, categorical features such as “Gender”, “Contract”, and “Payment Method” were converted into numerical representations using encoding techniques. This transformation was necessary because most machine learning algorithms require numerical input to function correctly.

In addition, the target variable “Churn” was converted into a binary format (0 for non-churn and 1 for churn), enabling it to be used in classification models. Care was also taken to maintain consistency in feature formats and avoid introducing bias during transformation.

Overall, feature engineering ensured that the dataset was structured, consistent, and suitable for model training, forming a strong foundation for subsequent analysis.

7. Feature Selection

Several features were removed during this stage due to their lack of predictive value or risk of data leakage. For example, variables such as “Churn Category” and “Churn Reason” directly indicate the outcome and would unfairly influence the model if included. Similarly, “Customer Status” was removed as it overlaps with the target variable.

Additionally, features such as “Customer ID” were excluded as they do not provide meaningful information for prediction. By removing such attributes, the model was allowed to focus on patterns that genuinely contribute to churn prediction.

This process resulted in a refined dataset with 31 features, improving both model efficiency and interpretability.

8. Model Implementation

The implementation phase involved training multiple machine learning models using the preprocessed dataset. Each model was trained under consistent conditions to ensure a fair comparison of performance.

Before training, the dataset was split into training and testing sets. The training data was further processed using SMOTE to address class imbalance, ensuring that the minority class (churn) was adequately represented. Feature scaling was applied to models that are sensitive to feature magnitudes, such as KNN, SVM, and Logistic Regression.

Each model was then trained using the processed training data, and predictions were generated on the test dataset. Care was taken to ensure that preprocessing steps such as scaling were applied consistently to both training and testing data to avoid discrepancies.

By following a structured implementation approach, the models were able to learn meaningful patterns from the data and produce reliable predictions.

9. Model Evaluation

Accuracy alone is not sufficient; recall for churn is critical.

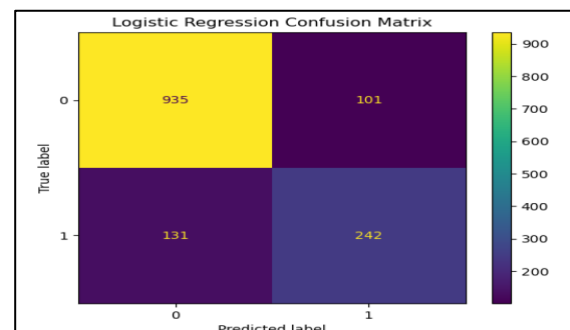


Figure 3: Confusion Matrix for Logistic Regression Model.

The confusion matrix in Figure 3 provides insights into model prediction and classification errors.

Multiple evaluation metrics were used to assess model performance. While accuracy provides an overall measure of correctness, it may not be sufficient in imbalanced datasets. Therefore, precision, recall, and F1-score were also considered to provide a more comprehensive evaluation.

In the context of churn prediction, recall is particularly important as it measures the model's ability to correctly identify customers who are likely to churn. Missing such customers can result in significant business losses.

10. Model Optimisation

Model optimisation was performed to improve the performance and generalisation capability of the machine learning models. Hyperparameter tuning was carried out using GridSearchCV, which systematically searches for the best combination of parameters by evaluating multiple configurations through cross-validation.

Each model has specific hyperparameters that influence its behaviour. For example, Logistic Regression depends on regularisation strength, Decision Trees rely on depth and splitting criteria, and Random Forest involves the number of trees and their depth. By tuning these parameters, the models can be adjusted to better fit the data while avoiding overfitting.

A key observation from this process was that the accuracy of most models slightly decreased after tuning. This indicates that the initial models may have been overfitting the training data, while the tuned models provide a more realistic estimate of performance on unseen data.

For computationally intensive models such as SVM, a reduced parameter grid was used to balance performance with execution time. Overall, model optimisation played a vital role in improving the robustness and reliability of the results.

11. Feature Importance

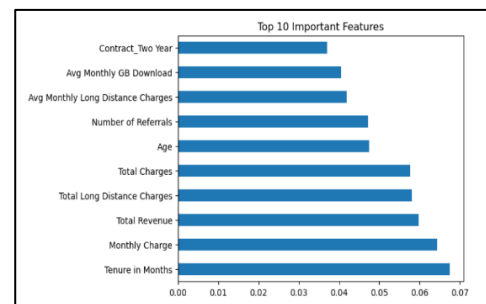


Figure 4: Top 10 Important Features from Random Forest Model

Figure 4 highlights the most influential features contributing to churn predictions.

Feature importance analysis was conducted to identify the most influential variables contributing to customer churn prediction. This was primarily performed using the Random Forest model, which provides a measure of feature importance based on how much each feature contributes to reducing impurity in decision trees.

The analysis revealed that features such as “Tenure in Months”, “Monthly Charges”, and “Total Revenue” were among the most significant predictors. These features are directly related to customer engagement and financial behaviour, making them highly relevant for churn prediction.

Additionally, service-related features such as contract type also played an important role, indicating that customers with certain types of contracts are more likely to remain with the service.

Understanding feature importance not only improves model interpretability but also provides valuable business insights. For example, companies can focus on improving customer experience for users with shorter tenures or high monthly charges to reduce churn rates.

12. Challenges and Solutions

During the development of this project, several practical challenges were encountered, which required iterative debugging and refinement of the machine learning pipeline.

One of the primary issues was a significant reduction in dataset size during preprocessing. Initially, the dataset contained over 7000 records, but incorrect handling of missing values led to unintended row deletions. This was resolved by carefully applying the appropriate imputation techniques instead of dropping rows, ensuring that the dataset retained sufficient information for modelling.

Another challenge involved handling class imbalance. The dataset contained a higher proportion of non-churn customers, which could bias the models. This issue was addressed using the Synthetic Minority Oversampling Technique (SMOTE), applied only to the training dataset to avoid data leakage.

While implementing SMOTE, compatibility issues were encountered due to version conflicts between libraries. This resulted in reinstalling and updating relevant packages to ensure smooth integration.

Hyperparameter tuning using GridSearchCV also presented computational challenges, particularly for the Support Vector Machine (SVM) model. The process was time-consuming due to the complexity of the algorithm and the increased dataset size after SMOTE. To address this, a reduced parameter grid was used along with parallel processing to optimise execution time.

Additionally, minor coding errors such as incorrect parameter names and undefined variables (e.g., issues with train-test splits and feature scaling) were identified and corrected during debugging.

Overall, resolving these challenges improved both the robustness and reliability of the final

models, ensuring that the results were based on a well-structured and validated pipeline.

13. Results and Discussion

	Model	Before Tuning	After Tuning
0	Logistic Regression	0.83	0.76
1	Decision Tree	0.80	0.79
2	Random Forest Model	0.82	0.81
3	KNN	0.72	0.65
4	SVM	0.75	0.73

Figure 6: Accuracy results before and After Tuning.

The results indicate that Random Forest achieved the highest accuracy after hyperparameter tuning, making it the most effective model for this dataset. Its ensemble nature allows it to capture complex relationships while reducing the risk of overfitting.

A notable observation is that all models experienced a slight decrease in accuracy after hyperparameter tuning. While this may appear counterintuitive, it actually suggests that the initial models were overfitting the training data. GridSearchCV, combined with cross-validation, provides a more reliable estimate of model performance on unseen data.

Logistic Regression also demonstrated strong performance, indicating that the dataset contains relatively linear relationships. In contrast, KNN showed the lowest performance, likely due to its sensitivity to feature scaling and data distribution.

Overall, the results highlight the importance of model selection, proper tuning, and evaluation strategies in building reliable machine learning systems.

14. Ethical, Legal and Professional Considerations

Machine learning models applied to customer data must ensure privacy and data protection. Sensitive customer information should be handled securely and comply with regulations such as GDPR.

Bias in data can also affect predictions. For example, demographic features may unintentionally introduce unfair bias. It is important to ensure that models are fair and transparent.

From a professional perspective, models should be interpretable and reliable before deployment in real-world systems.

15. Future Work and Conclusion

In conclusion, this study successfully implemented a comprehensive machine learning pipeline for predicting customer churn. The results demonstrate that ensemble methods such as Random Forest are particularly effective for this task, achieving the highest accuracy among all models.

The project also highlights the importance of addressing real-world challenges such as class imbalance, data preprocessing, and computational constraints. By applying techniques such as SMOTE, feature scaling, and hyperparameter tuning, the models were able to achieve improved generalisation.

Future work could explore more advanced techniques such as dimensionality reduction using PCA, deep learning models, and deployment of the solution in a real-time environment. Overall, this study provides a strong foundation for applying machine learning techniques to business-critical problems such as customer churn prediction.

16. References

- [1] Kaggle, "Telecom Customer Churn Dataset", Available: <https://www.kaggle.com/>
- [2] Scikit-learn Documentation, Available: <https://scikit-learn.org/>
- [3] Chawla, N. V. et al., "SMOTE: Synthetic Minority Over-sampling Technique", Journal of AI Research, 2002.
- [4] Han, J., Kamber, M., "Data Mining: Concepts and Techniques", Morgan Kaufmann.

- [5] Géron, A., "Hands-On Machine Learning with Scikit-Learn", O'Reilly Media.